Student Code Online Review and Evaluation 2.0

Dorothy Ammons, Shamik Bera, Patrick Kelly, Rakan Alsharif

Faculty Advisor(s): Dr. Raghuveer Mohan, Dept. of Electrical Engineering and Computer Science, Florida Institute of Technology

Goals

- The goal of this second rendition to S.C.O.R.E is to enhance the application by adding features such as:
  - Cheat detections
  - Roster imports
  - Grade exports
  - Custom rubrics

Motivations

- Our motivations lie in resolving areas that are lacking in the previous version, including but not limited to:
  - Ease of development
  - Tools related to Canvas implementation
  - Tools to customize the grading process: rubrics, cheat detections, custom verifiers

Implementation

- Web app: React + Node.js + Vite
- Database: Firestore + Google Cloud
- Backend: Flask + Python
- Deployment: PyInstaller

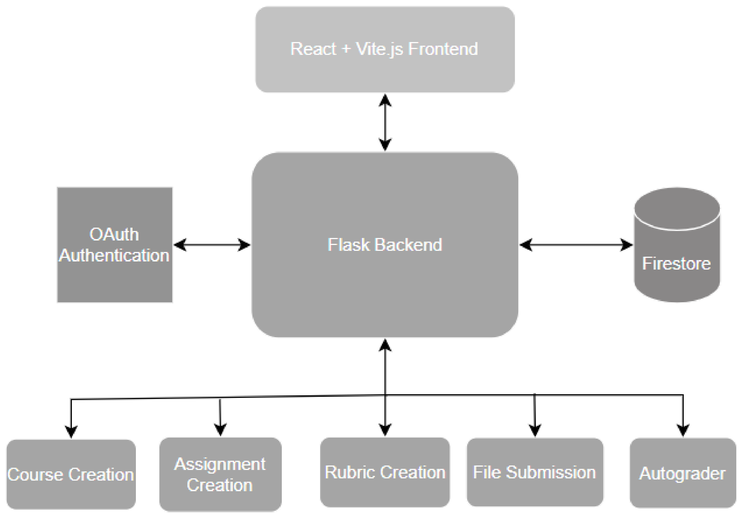
Limitations

- Selective coding languages supported
- Restricted threads allowed for autograding
- Constrained space for stored submission files
- Chrome required
- Windows or Linux operating systems required

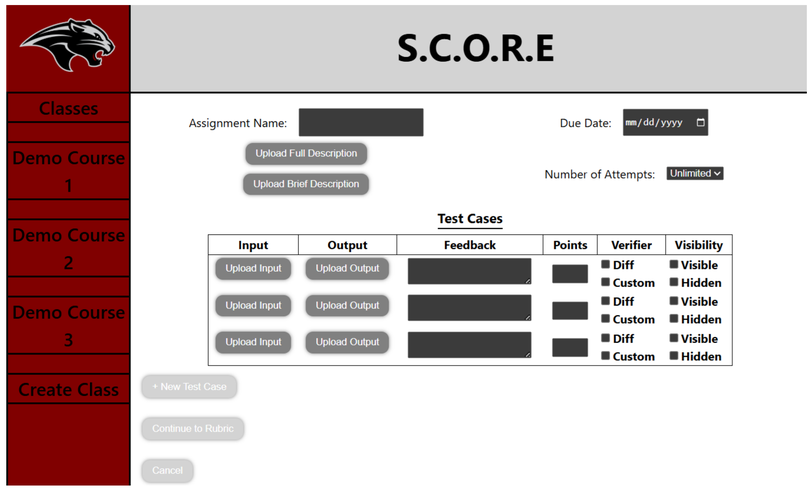
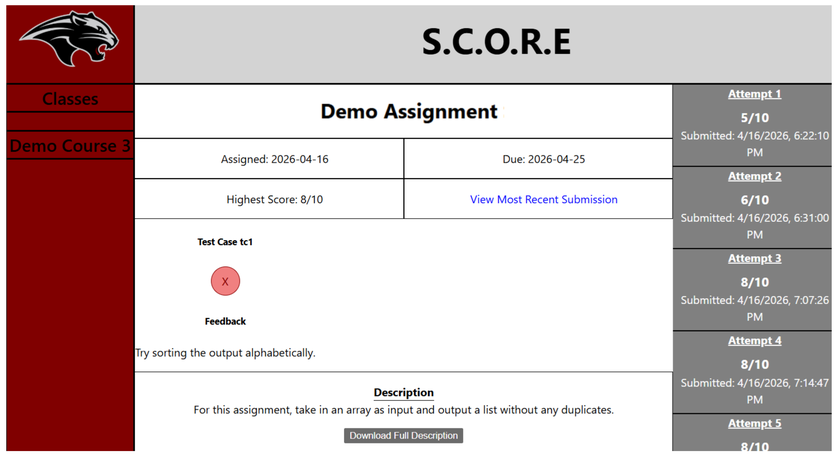
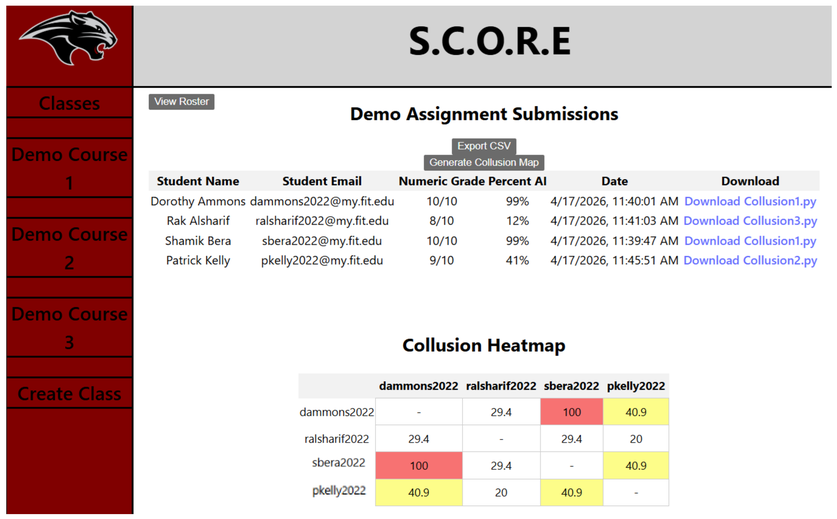
Features

- Google OAuth login
- AI and collusion detection
- Automated rubric based grading:
  - Late penalties
  - Point allocations
- Canvas style roster importing
- Canvas style grade exporting
- Custom code verification
- Testcase feedback

System Architecture



Future Improvements

- Hosting provided by Florida Institute of Technology
- Improved AI detection and visuals
- Improved collusion detection and visuals
- Command line interface
- Support for Mac operating systems

Interface